

WORKSHEET 7

Establishing Test Validity and Reliability

Group Work

Group 1:

Cantillo, Prince

Gardoce, Rose Marianne Louis M.

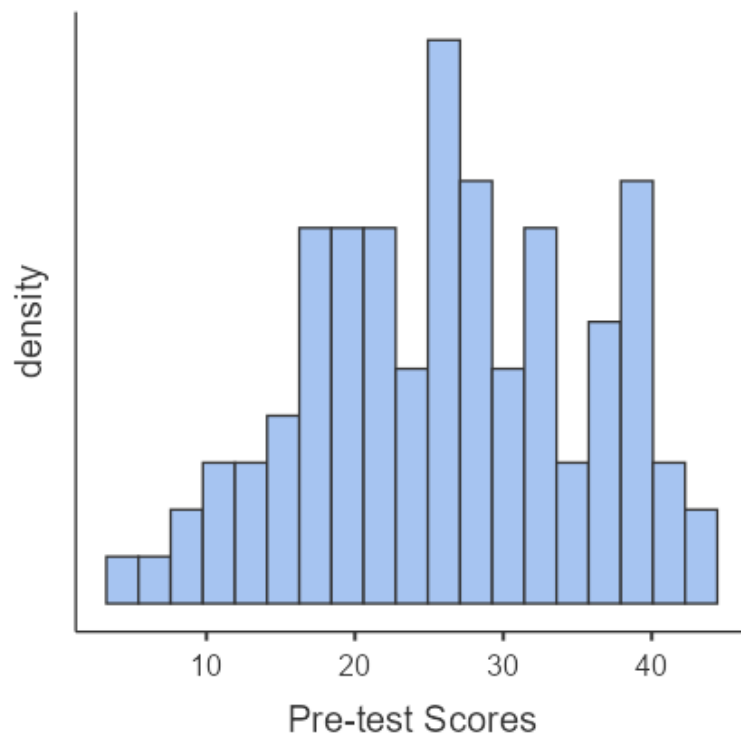
Jaramillo, Arnold Daniel

Masculino, Cherry

Instructions:

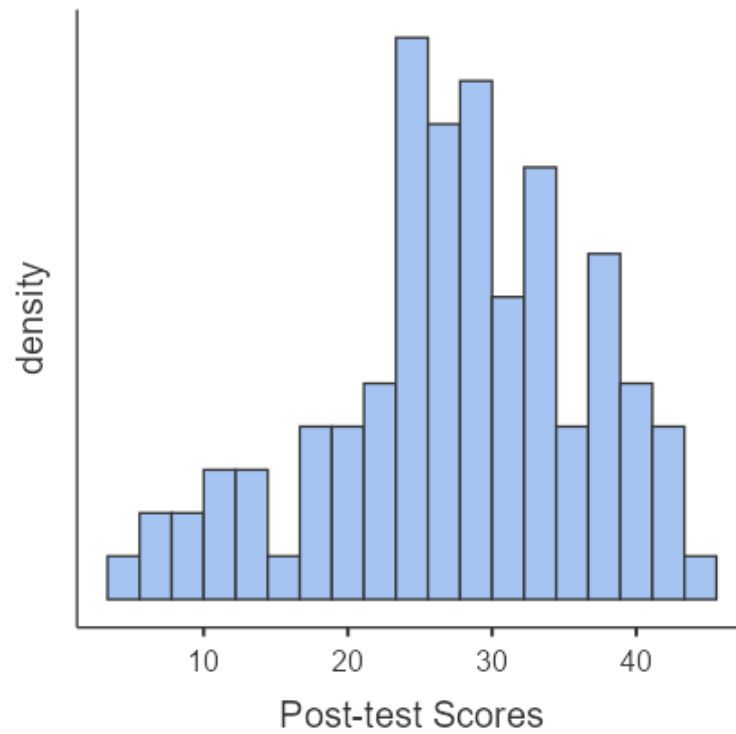
1. Summarize the two datasets.
 - a. Histogram

Pre-test Scores



Pre-test scores show a slightly negatively skewed, continuous distribution across the class. Based on the pattern of scores, most students achieved mid-to-high scores, with the highest concentration of students scoring between 25 and 27. Only a small number of students scored below 15, while another noticeable cluster performed well between 38 and 40.

Post-test Scores



Post-test scores show a left-skewed distribution across the class. Based on the pattern of scores, the majority of students scored higher on the assessment, with the highest concentration of students scoring between 24 and 30. Only a small number of students scored below 15, while a notable portion of students scored in the higher range between 32 and 39.

b. Mean

Descriptives		
	Pre-test Scores	Post-test Scores
Mean	26.1	27.6
Median	26.0	28.0
Standard deviation	9.29	9.03
Variance	86.2	81.5
Range	39	40
Minimum	5	4
Maximum	44	44
50th percentile	26.0	28.0
75th percentile	33.0	34.0
100th percentile	44.0	44.0

Pre-test and post-test scores both show consistent average performance with a slight improvement following the post-test. Based on the descriptive statistics, the mean score increased from 26.1 in the pre-test to 27.6 in the post-test, while the median score rose from 26.0 to 28.0. Both assessments exhibit a similar degree of variability, with standard deviations of 9.29 and 9.03 respectively, and identical maximum scores of 44.

c. Standard deviation

The standard deviation remained virtually stable between the pre-test (9.29) and post-test (9.03), showing that the overall spread of student scores did not noticeably change following instruction. A standard deviation of around 9 points relative to the means (26.1 and 27.6) reflects a moderate-to-high degree of variability, indicating that student performance was widely dispersed across the score range rather than tightly clustered around the average.

d. Minimum and maximum scores

The minimum score slightly decreased from 5 in the pre-test to 4 in the post-test, while the maximum score remained identical at 44 across both assessments. This indicates that the total range of performance spanned across nearly 40 points in both tests.

e. Percentiles

- 50th Percentile (Median): Increased from 26.0 to 28.0, meaning half of the students scored 28 or higher on the post-test compared to 26 on the pre-test.
- 75th Percentile: Rose from 33.0 to 34.0, showing that the top 25% of students scored at or above 34 in the post-test compared to 33 in the pre-test.
- 100th Percentile: Remained unchanged at 44.0, indicating the highest individual score achieved across both tests stayed the same.

2. Compare the two datasets using the organized summary results.

Overall, the students showed a slight improvement on Test 2 compared to Test 1. The mean score increased from 26.11 to 27.58, and the median shifted upward from 26 to 28. Despite this increase in the average score, the spread of the data remained highly consistent; both tests had nearly identical standard deviations (9.29 vs. 9.03) and the same maximum score limit (44).

3. Generate the scatterplots. Interpret the scatter plot together with the computed correlation of the two tests.

Pre-test and post-test scores have a strong positive linear relationship. The scatter plot demonstrates that data points tightly cluster around the upward-sloping trendline, indicating that students who scored higher on the pre-test consistently scored higher on the post-test, while those who scored lower on the pre-test tended to score lower on the post-test as well.

CORRELATION: Strong positive ($r \approx .91$)

4. Each group member will randomly choose three (3) students and compare their performance in both tests in terms of their percentile ranks.

Gardoce, Rose Marianne Louis					
Student ID	Pre-test Score	Pre-test Percentile Rank Range	Post-test Score	Post-test Percentile Rank Range	Performance Interpretation Student_055
Student_055	37	86th percentile	39	91st percentile	In the pre-test, Student 055 performed better than approximately 86% of the group, compared with approximately 91% in the post-test. Both scores were above the group's 50th-percentile scores of 26 for the pre-test and 28 for the post-test. This also shows that Student 055 performed better in post-test.
Student_063	13	9th percentile	12	8th percentile	Student 063 performed well or better than approximately 9% of the group in the pre-test but only 8% in the post-test. Both scores were below the group's 50th-percentile scores of 26 for the pre-test and 28 for the post-test. This also shows that Student 063 performed better in pre-test.
Student_074	41	96th percentile	44	100th percentile	Student 074 performed well or better than approximately 96% of the group in the pre-test and 100% in the post-test. Both scores were above the group's 50th-percentile scores of 26 for the pre-test and 28 for the post-test. This also shows that Student 074 performed better in post-test.

Masculino, Cherrie Masculino

Student ID	Pre-test Score	Pre-test Percentile Rank Range	Post-test Score	Post-test Percentile Rank Range	Performance Interpretation Student_055
Student_007	39	92 nd percentile	38	88 th percentile	Student_007 performed better than approximately 92% of the class in the pre-test and 88% in the post-test. Both scores are above the group's 50th percentile (26 for pre-test and 28 for post-test). However, there is a decrease in the post-test score, though the student still performed above the 50th percentile.
Student_019	15	12 th percentile	23	24 th percentile	Student_019 performed better than approximately 12% of the class in the pre-test and 24% in the post-test. Both scores are below the group's 50th percentile (26 for pre-test and 28 for post-test). However, there is an increase in the post-test score, indicating improvement.
Student_026	32	72 nd percentile	29	54 th percentile	Student_026 performed better than approximately 72% of the class in the pre-test and 54% in the post-test. Both scores are above the group's 50th percentile (26 for pre-test and 28 for post-test). However, there is a decrease in the post-test score, though the student still performed above the 50th percentile.

Cantillo, Prince Nimrod C.					
Student ID	Pre-test Score	Pre-test Percentile Rank Range	Post-test Score	Post-test Percentile Rank Range	Performance Interpretation Students
Student_013	32	69.7 th percentile	28	49.5 th percentile	This student experienced a significant drop in performance. Even though they started at the top edge of the 70th percentile, losing 4 points on the second test caused their class rank to plummet down to the median (50th percentile).
Student_060	36	80.8 th percentile	40	91.9 th percentile	This student demonstrated exceptional improvement. They started in the 80th

					percentile on Test 1, but a 4-point increase on Test 2 successfully pushed them into the elite 90th+ percentile bracket of the class.
Student_048	39	90.9 th percentile	38	85.9 th percentile	This student is a consistently high achiever but experienced a slight regression. Losing just a single point (from 39 to 38) was enough to knock them out of the 90th percentile and down into the 85th, showing how highly competitive the top end of this class distribution is.

Jaramillo, Arnold Daniel B.					
Student ID	Pre-test Score	Pre-test Percentile Rank Range	Post-test Score	Post-test Percentile Rank Range	Performance Interpretation Students
Student_97	33	75 th percentile	33	71 st percentile	Despite retaining their previous score, the student's percentile rank has dropped from 75 th to 71 st . This is caused by 4 other students that were once under the 75 th percentile now having a higher score than the student.
Student_93	21	32 nd percentile	23	24 th percentile	This student improved their score, but their ranking actually went down. The improvement wasn't as significant as other students that were below the 32 nd percentile in the pretest. There are at least 8 students that improved there scores.
Student_08	40	95 th percentile	37	84 th percentile	The student had a slight regression in their score however their percentile rank dropped drastically which means that more students are reaching scores above 37.

5. Submit the group's output in WORD/PDF format with filename format: **Group#-WS7.**
6. Due on September 23 at 5:30 pm (before our class).